HOOK

Your coding agent isn't dumb. Your prompt is.

Two deep technical threads this week landed on the same conclusion.

WHAT'S HAPPENING

Large codebases are exposing a real pattern

Developers running agents on big, messy real-world projects keep hitting the same wall.

- One deep-dive: a massive decompilation experiment at scale
- One essay: dissecting why coding agents underperform
- Both converge on the same root cause
- Neither blames the model's raw intelligence

THE REAL CAUSE

It's not intelligence. It's structure.

Agents don't fail because they can't reason. They fail because nothing checks their reasoning.

- No checkpoints to catch drift early
- No validation between steps
- Scope too wide for the agent to stay grounded
- One giant prompt instead of a managed process

MECHANISM

Treat agent work like a managed process

The same discipline you'd apply to a junior engineer or a CI pipeline.

- Break the task into narrow, verifiable steps
- Define what 'correct' looks like before it runs
- Insert a validation step after each stage
- Let the agent fail small, not fail big

WHY IT WORKS

Feedback loops are the missing ingredient

Open-ended prompts give the agent room to drift without anyone noticing.

- Checkpoints catch errors before they compound
- Narrow scope reduces ambiguity in interpretation
- Validation turns output into a decision, not a guess
- Smaller loops make mistakes cheap to catch

WHAT TO CHANGE

Stop asking agents to do everything at once

The fix isn't a smarter model. It's a smarter process around it.

- Define the checkpoint before you define the prompt
- Ask: what would tell me this step actually worked?
- Keep each task scoped to one verifiable outcome
- Review outputs as stages, not as a finished product

TAKEAWAY

Architecture beats model choice, every time

This is the same lesson that applies to any agentic system, not just coding agents.

- Good structure makes an average model reliable
- Poor structure makes a great model unpredictable
- Human-in-the-loop checkpoints aren't a limitation
- They're what makes autonomy usable in production

CTA

Where is your agent working without a checkpoint?

Curious how this shows up in your own workflows.